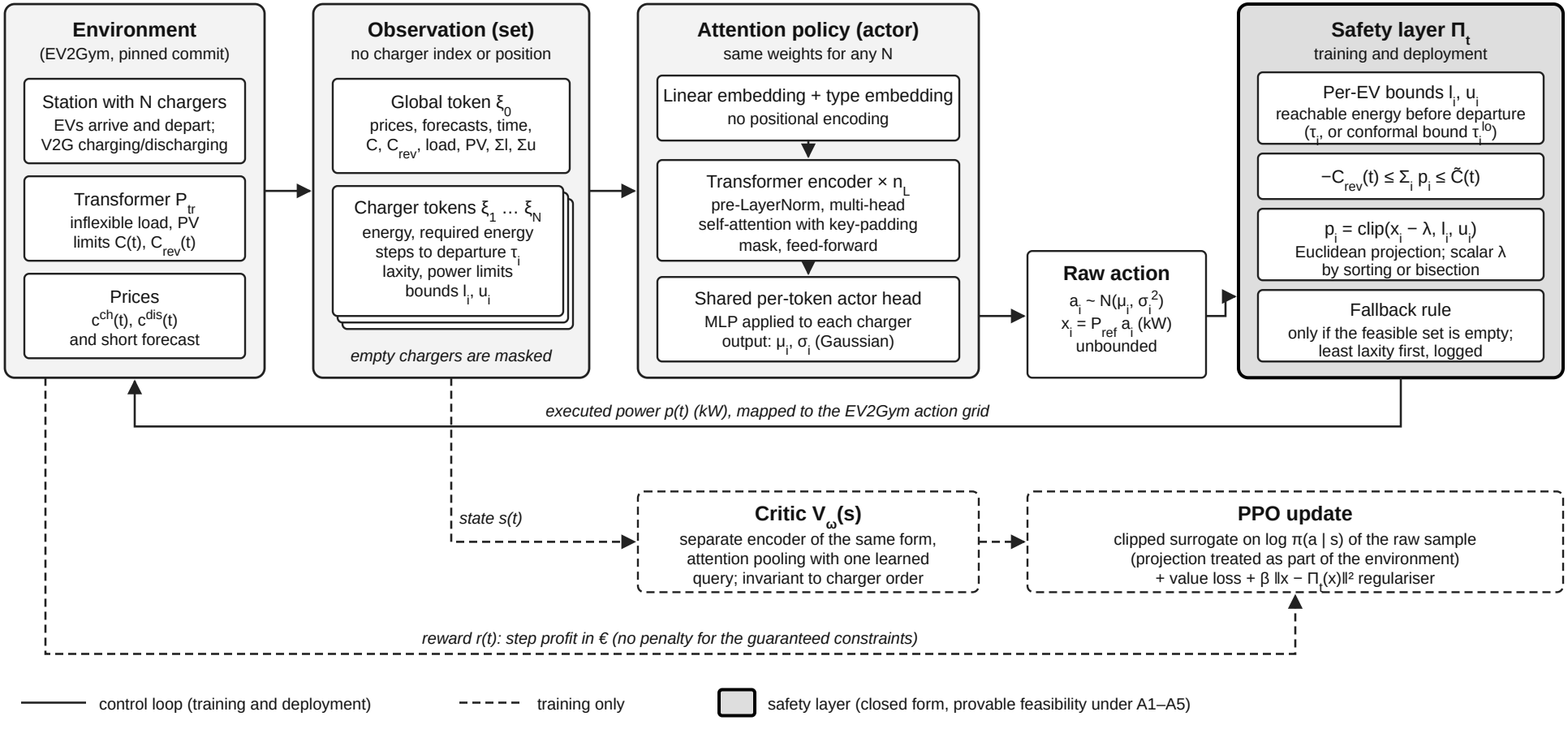
Environment
(EV2Gym, pinned commit)
Station with N chargers
EVs arrive and depart;
V2G charging/discharging
Transformer P_{tr}
inflexible load, PV
limits C(t), C_{rev}(t)
Prices
c^{ch}(t), c^{dis}(t)
and short forecast
Observation (set)
no charger index or position
Global token ξ_0
prices, forecasts, time,
C, C_{rev}, load, PV, Σl, Σu
Charger tokens ξ_1 … ξ_N
energy, required energy
steps to departure τ_i
laxity, power limits
bounds l_i, u_i
empty chargers are masked
Attention policy (actor)
same weights for any N
Linear embedding + type embedding
no positional encoding
Transformer encoder × n_L
pre-LayerNorm, multi-head
self-attention with key-padding
mask, feed-forward
Shared per-token actor head
MLP applied to each charger
output: μ_i, σ_i (Gaussian)
Raw action
$a_i \sim N(\mu_i, \sigma_i^2)$
$x_i = P_{ref}\, a_i$ (kW)
unbounded
Safety layer Π_t
training and deployment
Per-EV bounds l_i, u_i
reachable energy before departure
(τ_i, or conformal bound τ_i^{lo})
$-C_{rev}(t) \le \Sigma_i\, p_i \le \tilde{C}(t)$
$p_i = \mathrm{clip}(x_i - \lambda, l_i, u_i)$
Euclidean projection; scalar λ
by sorting or bisection
Fallback rule
only if the feasible set is empty;
least laxity first, logged
executed power p(t) (kW), mapped to the EV2Gym action grid
state s(t)
Critic V_ω(s)
separate encoder of the same form,
attention pooling with one learned
query; invariant to charger order
PPO update
clipped surrogate on log π(a | s) of the raw sample
(projection treated as part of the environment)
+ value loss + β ‖x − Π_t(x)‖² regulariser
reward r(t): step profit in € (no penalty for the guaranteed constraints)
control loop (training and deployment)
training only
safety layer (closed form, provable feasibility under A1–A5)